



OPEN Interpretable deep learning models for independent fertilizer and crop recommendation

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The integration of Artificial Intelligence (AI) and Internet of Things (IoT) technologies is fundamentally transforming agriculture. This integration enables the real-time collection and analysis of critical data such as soil nutrient levels, temperature, humidity, and climatic conditions. A data-driven approach allows for independent fertilizer and crop recommendations, maximizing yield and promoting resource efficiency. Soil properties play a crucial role in crop growth, facilitating the absorption of essential nutrients like Nitrogen (N), Phosphorus (P), and Potassium (K). However, traditional farming practices often lead to nutrient inefficiency and soil health degradation, negatively impacting productivity. While Machine Learning (ML) techniques have emerged for recommending fertilizers and crops, they frequently encounter issues such as inadequate feature selection and class imbalance. These limitations hinder their ability to accurately model the intricate relationships among environmental factors, soil conditions, and crop nutrient requirements. To address these challenges, this study presents a unified smart recommendation system that utilizes TabNet, a deep learning architecture specifically designed for tabular data. The novelty of this work lies in leveraging TabNet's attention-driven learning to directly discover important patterns from preprocessed IoT-enabled agricultural data for accurate and interpretable crop and fertilizer classifications, without relying on prior feature selection. To enhance transparency, SHapley Additive exPlanations (SHAP) is applied at the final stage to provide post hoc interpretability, allowing stakeholders to understand the model's reasoning behind each recommendation. Evaluated with the Crop and Fertilizer Dataset from Western Maharashtra, TabNet achieves impressive classification accuracies of 95.24% for fertilizer recommendations and 96.21% for crop recommendations, outperforming conventional classifier and existing approach. The study employs robust preprocessing techniques, such as iterative imputation and the Synthetic Minority Oversampling Technique (SMOTE), to ensure data quality and class balance. The model's performance and generalizability were rigorously assessed using fivefold cross-validation, consistently maintaining these high accuracy levels across folds.

Keywords Internet of Things, Smart farming, Fertilizer recommendation, Crop recommendation, Deep learning, Tabnet, Shapley Additive exPlanations (SHAP), Explainable artificial intelligence

The agriculture industry is an essential contributor to a nation's economic prosperity. Preserving soil quality is essential for effective crop production. Utilizing advanced technologies for planning, analysis, and production management is essential for improving agricultural productivity, encompassing soil health, plant nutrition, and water quality^{1,2}. As individuals increasingly prioritize health, the cultivation of high-quality crops is essential for contemporary society. Agriculturists use pesticides and incorporate nutrients into the soil to achieve optimal yield. Pesticides are toxic agents employed to exterminate pests, weeds, fungi, and similar organisms, encompassing herbicides, fungicides, and insecticides³. The implementation of IoT in smart farming streamlines agricultural processes by facilitating precise seed sowing and efficient fertilizer application, resulting in increased crop yields⁴. As the global food demand continues to grow, there is a rising necessity for automated, intelligent, and scalable technologies capable of accurately assessing soil and crop conditions and advising on the most appropriate fertilizers⁵. An innovative approach for improving agricultural operations is precision agriculture that uses data-driven decision-making to improve agriculture. Fertilizer/crop recommendation is crucial since it boosts agricultural productivity and reduces resource waste and environmental effect. Traditional fertilizer/crop recommendation methods involve soil testing and heuristic-based procedures, which are laborious, time-consuming, and error-prone due to soil composition and environmental conditions⁶. Integration of smart

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farming technologies, especially Internet of Things (IoT) in AI based fertilizer/crop recommendation systems can lead advanced revolution in this direction. IoT sensors continuously monitor the soil health by collecting data in real time on the soil moisture, pH, temperature, and nutrient levels⁷. The integration enhances the scalability of AI-based farming solutions and reduces reliance on conventional soil testing methods⁸.

Now, recent advancements in Artificial Intelligence (AI) and deep learning for data-driven solutions have transformed agricultural decision-making. The black box nature of deep learning systems has limited its agricultural uptake even though it demonstrates effective performance in farming operations. Technology limitations in deep learning models make it difficult for farmers and agronomists to understand prediction reasoning because these models contain complex non-linear function algorithms⁹. Users gain better understanding of model behavior and feature importance by applying attention mechanisms alongside TabNet, SHAP, and LIME (Local Interpretable Model-Agnostic Explanations)¹⁰. The trust and acceptance problems stemming from explainability shortages create particular challenges for important decision domains particularly shown through fertilizer/crop application systems. The development of explainable AI (XAI) approaches provides solutions to this issue through decision-making explanation techniques¹¹.

The implementation of XAI improves AI-powered fertilizer/crop recommendation system transparency, so farmers and specialists will build greater confidence in the solutions¹². Systems that recommend fertilizers/crop gain advantages from combining deep learning with XAI because it enhances their performance quality along with their reliability and adaptability. Through AI model interpretability stakeholders obtain the ability to see how input data affects recommendations while detecting biases and implementing recommendations through domain knowledge¹³. Deep learning algorithms can produce more accurate and context-sensitive fertilizer/crop recommendations by combining this data with satellite imaging and remote sensing methods, thus enhancing nutrient management for diverse soil types and crops. Sustainable agriculture requires humans to pair AI systems with their expertise to create suggestions that help achieve environmental sustainability targets and respect agronomic standards¹⁴. The goal of this research aims to determine how deep learning and XAI can create sustainable fertilizer/crop recommendation systems that maintain reliability and scalability and offer interpretability to users. The proposed approach utilizes TabNet along with XAI approaches to enhance nutrient/crop management, precision agriculture methods and sustainably produce food¹⁵. Deep learning techniques such as Transformer-based architectures, Convolutional Neural Networks (CNNs), and Recurrent Neural Networks (RNNs) have been shown to be particularly well suited to handle complex and high-dimensional data in agriculture applications such as soil nutrient assessment, disease detection, and yield estimation¹⁶. TabNet is a state-of-the-art deep learning architecture, tailored for tabular data, that has shown promise in fertilizer/crop recommendation. Its ability to capture complex interactions between features while remaining interpretable is the main reason for that. Unlike traditional machine learning models, which require extensive feature engineering, TabNet can directly learn feature representations, making it appropriate for heterogeneous agricultural datasets^{17,18}.

TabNet has selected as the core classification model due to its unique architecture tailored for tabular data. Unlike conventional deep learning models that often underperform on tabular datasets, TabNet leverages sequential attention and sparse feature selection to focus on the most relevant input features during each decision step. This allows it to retain interpretability and efficiency—key requirements for agricultural decision-making systems. While other models like NODE (Neural Oblivious Decision Ensembles)¹⁹ and FT-Transformer²⁰ have shown competitive performance, TabNet offers the advantage of built-in explainability and superior training speed on structured datasets. These properties made it well-suited for the proposed approach of fertilizer or crop recommendation using static agronomic features.

As a result, the primary goal of this study is to use intelligent technology to recommend effective fertilizer/crop for soil, specifically addressing the key challenges in fertilizer/crop recommendation, namely class imbalance, poor feature selection, and the complexity of soil-climate-crop relationships. Traditional machine learning algorithms suffer from interpretability issues and inadequate feature representation and therefore, the proposed work presents a more sophisticated methodology to tackle the problem at hand. To enhance the interpretability of the model, the proposed system employs SHAP explainability and Tabnet's interpretability, iterative imputation for missing values, SMOTE for balancing the dataset. Moreover, TabNet is an interpreted deep learning model specifically designed for tabular data that uses attention-based learning and sequential feature selection to learn the correlations among nutrient levels, environmental variables, and soil properties. Hence, TabNet outperforms existing models in terms of predictive ability and interpretability according to experimental data. This research has significant potential for precision agriculture as it provides a precise, interpretable, and computationally efficient tool, allowing agricultural experts to make data-driven fertilizer/crop decisions that augment crop yield and promote sustainable soil management.

The paper is organized as follows: Section "Literature review" examines pertinent literature on smart agriculture, independent fertilizer and crop recommendations. Section "Proposed methodology" delineates the proposed approach for smart fertilizer/crop recommendations, whereas Section "Evaluation and discussion" describes the experimental configuration and evaluates the results. Section "Conclusion" ultimately finishes the work by discussing major findings and recommended avenues for future research.

Literature review

Smart farming gained significant innovation from the Internet of Things (IoT) because this technology allows precision agriculture together with automation systems and quick decision-making. Production optimization in IoT-based smart farming depends on the core components which include sensors with communication networks as well as data analytics functions. Climate control and real time environmental surveillance emerge as the essential applications for IoT in farming sector. Prashanth et al. conducted a comprehensive assessment of the wider adoption of IoT in the smart farming sector, extending beyond their initial research on temperature

regulation. They examined deep learning alongside IoT innovations relevant to agricultural production. The study confirmed that deep learning models, specifically CNN and MobileNet, are effective for detecting diseases and predicting yields. Additionally, the research indicated that integrating various optimization approaches could improve IoT node placement, thereby increasing coverage area and reducing energy consumption^{4,21}. Castaneda-Miranda et al. designed an intelligent frost control system featuring FES and ANN technology for automated greenhouse anti-frost watering through temperature prediction capabilities. The system proves effective for predicting temperature variations across seasons since it relies on Internet of Things-connected weather stations to monitor weather conditions in real-time⁵. Javaid et al. performed research about AI technologies which show rising promise for agricultural transformation particularly under climate change conditions. These technologies optimize resource management, improve meteorological predictions, and promote sustainable agricultural practices, ultimately leading to increased crop yields. Their study showed that large climatic datasets utilized for strengthening farming systems against environmental changes by AI driven real-time decision-making⁶.

Klerkx et al. analyzed that multidisciplinary research becomes vital to address the problems of adoption and ethics as well as policy frameworks in digital agriculture analysis. The study identified essential digital agriculture topics which included knowledge-sharing mechanisms together with power frameworks and technology implementation modes²². Wolfert et al. examined how big data modifies smart farming by analyzing its impact on farm management and operational processes and enterprise organization. Big data analytics transforms agricultural supply chains by giving predicted insights together with real-time operational enhancements²³. Liakos et al. studied how large data together with advanced computing systems impact the application of machine learning in agriculture. ML successfully manages crops, livestock, water resources and also controls soil quality. Three essential agricultural applications of high importance include yield prediction along with disease detection, weed detection and livestock monitoring. Their work illustrated ML-based solutions improve real time decision making and farming practices⁷. Tzounis et al. analyzed modern IoT movements in agriculture while focusing on how cloud and fog systems handle big agricultural data collections. The presentation identified data integration requirements for farm productivity enhancement while discussing IoT device interoperability problems²⁴. Youness et al. by introducing a smart irrigation system based on Internet of Things and machine learning. They used sensors to measure soil moisture, temperature, and rainfall, while KNN and neural networks scheduled watering. They prove that by using AI-driven IoT systems they can reduce water consumption while making sure that the plants are still able to grow to the fullest potential, which is the main premise of their finding⁸. Ray et al. discusses IoT-based solutions and presents several smart farming systems with sensors in various case studies. Their analysis also identified numerous technological challenges that need to be addressed to unlock the full potential of IoT in agriculture, such as network limitations and data security problems²⁵.

Ahmed et al. developed a soil fertilization recommendation system which utilizes an IGA to process time-dedicated sensor information for finding excellent crop management parameters. A neighborhood-based strategy within the model enables it to optimize parameter choices and produce superior yield outcomes. The system demonstrates improved ability in reporting soil health status while producing real-time fertilizer recommendations through sensor data evaluation²⁶. Thorat et al. presented an intelligent system for insecticide and fertilizer recommendations through combination of Transition Probability Function-Convolutional Neural Network (TPF-CNN). The system enables identification of pests and recommends insecticides through an API which combines NPK sensor technology to evaluate soil nutrients. The system provides fertilizer recommendations through its module by using real-time soil data to determine suitable nutrient levels which generates results in less than 80 s. Their methodology achieved better accuracy results when compared to standard machine learning approaches including K-Nearest Neighbors (KNN) and Support Vector Machine (SVM)⁹. Dey et al. crafted a machine learning system to guide agricultural and horticultural crop recommendations through combination of environmental factors, soil pH, and nutrient analysis. A research evaluation of the predictive performance involved five machine learning methods: Random Forest, Support Vector Machine, KNN, XGBoost, and Decision Tree. The XGBoost model reached the best accuracy performance thus showing its potential for AI-guided fertilizer recommendations and sustainable farming decisions making¹⁰. Helfer et al. conducted soil fertility evaluation through the model uses historical data while the prediction of crop productivity relies on climate event analysis. The research approach showed exceptional effectiveness in forecasting soil organic matter while simultaneously determining clay content through its strong correlation coefficients for fertility assessment in precision agriculture¹¹.

Bhattacharya et al. introduced the PCFRIMDS (Precision Crop and Fertilizer Recommendations Using Integrated Multimodal Data Fusion for Sustainable Agriculture) as an integrated system that performs multimodal data fusion for precision crop and fertilizer recommendations. This system combines NPK values with moisture content, image analysis, and geographical data using a Bidirectional Gated Recurrent Unit (BiGRU) along with an Ant Lion Fuzzy Principal Component Analyzer (ALFPCA). Their solution enhances agricultural recommendations and boosts operational speed to support sustainable farming¹². Sims et al. investigated the Australian plantation forest using satellite hyperspectral imaging to detect nutrient status and fertilizer needs. They found that EO-1/Hyperion data could improve nutritional forecast models, reducing the cost of nutritional assessments while increasing their accuracy. This study demonstrated that remote sensing technologies could enhance large-scale fertilizer decisions by analyzing soil and vegetation data¹³. Khan et al. developed an IoT based fertilizer recommendation system which monitors soil fertility levels in real time. The recommendation system uses Support Vector Machine (SVM), logistic regression (LR), K-Nearest Neighbor (KNN), and Gaussian naive Bayes (GNB) machine learning models to deliver accurate fertilizer results. Each time the GNB model was used it showed the best accuracy level while both enhancing real-time nutrient application efficiency and supporting precision farming techniques¹⁴. Islam et al. developed soil nutrient monitoring and crop recommendations in IoT systems use an ML framework. Multiple sensors measure real-time soil conditions combined with environmental factors so machine learning algorithms process this data to determine precise

fertilizer recommendations. Through this system farms gain higher production levels while using resources more efficiently and achieving data-based agricultural choices¹⁵. Jeon et al. used MIR spectroscopy to forecast soil properties essential for fertilizer recommendations in South Korea. Their research showed that GLMs outperformed XGBoost in both insurance applications and predictive domains. Additionally, multilayer TabNet architectures generated precise predictions in soil analysis while providing clear explanations, even though they required high computational operations²⁷.

Auzzas et al. proposed an adaptive deep neural network that uses hyperspectral data to predict various soil characteristics. The model effectively adapts to spectral libraries and identifies key spectral bands, resulting in better outcomes than state-of-the-art methods for predicting soil properties such as pH, organic carbon, nitrogen, and phosphorus. This analytical approach achieves higher R^2 benchmark datasets and improved prediction accuracy²⁸. Beg et al. evaluated the effects of plot-specific and government-recommended fertilizer guidelines through field-based experimentation in Bangladesh. The program fails to influence fertilizer usage except for TSP phosphate fertilizer, where a decline in usage results in production losses. According to the research findings, the implementation of data-driven recommendations heavily relies on farmer perception²⁹. Asrat et al. examined spectral soil analysis to develop fertilizer recommendations for East African maize fields. The study produced predictive models for soil properties through analysis by different spectrometers and Partial Least Squares Regression (PLSR). The Quantitative Evaluation of the Fertility of Tropical Soils (QUEFTS) model included these predictions for delivering customized fertilizer suggestions at particular sites. Research found spectral technology works properly for performing large-scale affordable soil investigation³⁰. Ekatherina et al. established an off-take model to recommend fertilizers for cocoa production in West Africa. The research team used information from 195 on-farm trials as they evaluated model recommendations against official national standards. The results show that off-take model strategies result in better cocoa crop productivity and monetarily efficient production but higher fertilizer costs may restrict their adoption³¹. Zenghui Jia et al. designed a multi-stage recommendation system for spring maize fertilizer using narrowband vegetation indices. They assessed multiple growth periods with drone-operated hyperspectral imagery to analyze nitrogen status and grain yield performance. Their long-term NPK fertilizer experiment evaluated topdressing needs based on identified vegetation indices. The results showed that applying 180 kg ha⁻¹ of nitrogen maximized grain output and created an economic fertilization strategy to enhance precision farming³².

Schut et al. identified the soil analysis method and field sampling procedures, revealing that prediction errors impact the precision of recommendations created for individual fields. They determined that developing exact field-specific recommendations relying on an individual composite soil sample does not work because soil nutrient evaluation methods contain built-in variability³³. Wenjie Yang et al. created an optimized NPK fertilizer recommendation method which incorporated topsoil available nutrient thresholds for wheat cultivation in drylands of Chinese territories. Experimental findings validated that their recommended fertilization strategy cut down N and P inputs by 17.5% and 43.5% respectively and still maintained yield quality. The approach achieved higher profitability for farmers while improving the way they used fertilizer³⁴. Wilhelm et al. utilized Random Forest and Support Vector Machines machine learning models to examine how soil microbiome data could forecast soil health metrics. Their analysis reached an R^2 value of ~0.8 which showed how well the microbiome could predict soil texture alongside tillage status and overall ground health status³⁵.

Yeshashree Mahale and colleagues developed a Crop Recommendation System (CRS) using machine learning to support farmers in Maharashtra, India. Their research created a predictive framework that utilized Long Short-Term Memory (LSTM) networks and Random Forest algorithms, achieving an accuracy of 92%. This system offers weather forecasts to help farmers choose suitable crops. Additionally, it features a user-friendly graphical interface and a crop calendar to assist farmers in making informed agricultural decisions³⁶. Piccoli et al. constructed a deep neural network which analyzes hyperspectral data to determine soil properties. The research group obtained a general prediction accuracy of 0.75 R^2 which included pH_{CaCl2} and CaCO₃ as variables with R^2 values surpassing 0.90. Techniques based on deep learning enable the improvement of soil analysis through scalable processes³⁷. Yang et al. developed optimized NPK fertilizer suggestions that used dryland wheat soil nutrient levels in China. The proposed fertilization model reduced both nitrogen inputs by 17.5% and phosphorus inputs by 43.5% while maintaining yield levels according to the study's findings³⁴. Dawid Skrzypczak et al. evaluated cellulose-based fertilizers as environmentally friendly replacements for chemical-based fertilizers. They showed improved soil structure together with active microbes and increased nutrient retention which bolstered sustainable agriculture through enhanced soil wellness and higher crop productivity³⁸.

Deep learning is rapidly becoming a game changer in agriculture by providing the ability to do advanced data analysis and make deeper data-driven decisions. Deep learning models can handle complex datasets and have been shown to improve crop monitoring, soil fertility prediction, and pest detection. Andreas Kamilaris et al. examined deep learning agricultural applications through their survey which reviewed forty studies demonstrating deep learning techniques in addressing agricultural problems. They established that artificial neural networks exceed standard photographic processing systems by performing accuracy testing and operational efficiency tests^{16,39}. Tommaso Adamo et al. developed an AI-based microservices platform that optimizes irrigation systems and incorporates fertigation functions for smart agriculture applications. This integrated IoT platform utilizes deep learning models to provide real-time decision-making, which helps minimize water usage and fertilizer loss while enhancing overall crop production⁴⁰. Sujatha et al. reviewed machine learning and deep learning for soil fertility classification and fertilizer prediction. The review discussed the ability of ML and DL models to accurately classify soil fertility levels and thus use them for optimal fertilizer application. They also pointed out research gaps such as the dependence of expensive lab tests and the few research on low cost, high accuracy classification models⁴¹.

The deep learning model TabNet shows great capabilities to handle predictive modeling tasks for tabular data. McDonnell et al. examined how TabNet functions as an insurance predictive model while analyzing its

performance relative to XGBoost along with logistic regression models. The research findings illustrated TabNet generated interpretable results along with outstanding accuracy in risk assessment and claims forecasting and required higher computing power¹⁷. Arik and Pfister demonstrated that TabNet serves as an interpretable deep learning model which applies attention mechanisms to identify crucial features within tabular datasets. They conducted experiments and proved that TabNet surpassed standard tree-based models when used in different domains. The model exhibited different effectiveness across datasets because deep learning has restrictions when processing certain tabular data applications¹⁸. Ayan et al. utilized deep learning networks to classify agricultural pests by employing an ensemble of deep convolutional neural networks (CNNs). Multiple CNN architecture ensemble methods combined with genetic algorithms generated high accuracy results when validated across different datasets. Deep learning proves essential for early pest detection because it helps minimize crop damage as the study demonstrates^{42–44}.

Shams et al. developed XAI-CROP which serves as an explainable AI algorithm for crop recommendation to fulfill agricultural decision-making requirements through transparent methods. XAI-CROP used Gradient Boosting combined with Decision Trees and Random Forests which resulted in a 0.9412 Mean Squared Error and 0.94152 R-squared value while maintaining its capabilities for interpretation⁴⁵. Chen et al. illustrated that blockchain's secure ledger system enables perishable food monitoring which provides safety measures and minimizes hazards during supply chain operations. The participation of XAI provides stakeholders in the agricultural supply chain with understandable AI-driven decisions that establish trust through interpretations. The analysis of bibliometric data with VOS viewer helped researchers identify research dynamics, major authors and important publications in the field⁴⁶. Ryo et al. incorporated interpretability analysis for agricultural machine learning models in their research work. AI systems from traditional backgrounds operate in black box fashion which makes users unable to comprehend the predictive process. The study addressed this problem by using XAI methods on a dataset which analyzed no-tillage crop management to determine important factors that influence maize yield. Their work demonstrated through explainable machine learning methods that users gain access to various prediction rationale elements which enhances their trust in AI automation⁴⁷.

Recent advances in explainable AI (XAI) have enhanced transparency in agricultural decision-making, particularly in areas such as crop disease detection, yield prediction, and fertilizer recommendation. For example, Kamilaris et al. explored interpretable models for precision farming using SHAP values to understand nutrient-related recommendations⁴⁸, while Patel and Mehta integrated XAI techniques in decision trees for soil health-based fertilizer planning. These efforts underscore the need for transparent models in agriculture, motivating the use of TabNet's built-in feature masking to deliver interpretable fertilizer recommendations⁴⁹. Hosain et al. highlighted the importance of explainability in deep learning models, as these complex and intricate systems require explanations to enhance trust and accountability. Their study examined various methods of explainable artificial intelligence (XAI) and analyzed metrics to assess how interpretable these models are. They identified existing gaps in the current research and proposed future strategies to improve explainable AI approaches in deep learning systems, specifically aiming to apply them in sentiment analysis and other fields⁵⁰.

In summary, the increasing use of IoT in smart farming is revolutionizing agricultural practices, enhancing efficiency, sustainability, and productivity. However, challenges such as data privacy, interoperability, and infrastructure development continue to exist, necessitating further research and innovation. Agricultural productivity largely depends on soil fertility, as it directly influences crop growth and yield. The development of plants is significantly affected by the availability of essential nutrients like Nitrogen (N), Phosphorus (P), and Potassium (K). Researchers have explored various methods for generating effective fertilizer and crop recommendations by analyzing soil nutrients using intelligent algorithms and IoT systems. Deep learning and Explainable AI (XAI)-based systems for independent fertilizer and crop recommendation systems are revolutionizing modern agriculture by enhancing precision, sustainability, and efficiency. Leveraging IoT-enabled smart farming, machine learning models, and deep learning architectures, researchers are improving soil fertility assessment, optimizing fertilizer application, and guiding crop selection. While traditional machine learning models have shown promise in analyzing soil nutrients and recommending crop-specific fertilizers, they often face limitations in feature selection, class imbalance, and interpretability.

State-of-the-art interpretable deep learning approaches, such as TabNet, surpass traditional statistical methods by effectively modeling the complex interactions between soil, climate, and crop conditions for fertilizer and crop recommendations. However, many deep learning models operate as black boxes, making them difficult to interpret and challenging to apply in practical agricultural contexts. To address this issue, there has been a growing adoption of Explainable AI (XAI) techniques, such as SHAP and TabNet interpretability, which aim to improve model transparency and foster trust among farmers and agricultural stakeholders. This study further develops a TabNet-based framework for independent fertilizer and crop recommendations. This framework utilizes SMOTE and other advanced preprocessing techniques to enhance prediction accuracy. The results underscore the importance of combining deep learning with explainable AI in precision agriculture, fostering more intelligent, reliable, and sustainable farming practices for precise fertilizer and crop recommendations.

Proposed methodology

The proposed study develops a deep learning-based TabNet architecture that aims to provide independent recommendations for fertilizers and crops while enhancing interpretability, accuracy, and operational efficiency. By utilizing an attention mechanism and sequential feature selection technique, TabNet identifies exceptional relationships among soil, climate, and crop variables, enabling the generation of both precise and explainable recommendations. The transparency of decision-making processes is further enhanced through SHAP explainability and TabNet's inherent interpretability, which highlight the crucial features influencing these processes.

The performance of the proposed TabNet deep learning approach is evaluated by assessing the predictive model's accuracy, recall, precision, and F1-score. This evaluation demonstrates that TabNet effectively captures intricate relationships between features, with SHAP explainability contributing to the model's overall interpretability. The study's findings confirm that TabNet achieves superior predictive performance and enhanced explainability compared to existing systems. In this work, a single TabNet architecture is employed to separately predict fertilizers and crops. The proposed recommendation system enhances smart farming through the integration of deep learning, preprocessing, and explainability. Unlike traditional models, the interpretable deep learning framework of TabNet, combined with SMOTE-based data balancing, iterative imputation, and SHAP-based explainability, significantly improves prediction accuracy. A structured data pipeline analyzes the correlations between soil, crops, and fertilizers, generating independent recommendations based on soil and weather conditions, all supported by explainable AI insights.

The proposed methodology is intricately organized into several distinct layers, each playing a crucial role in the overall system architecture as illustrated by Fig. 1. At the forefront is the Data Collection Layer, which meticulously gathers a wide array of information pertaining to crops, soil conditions, weather patterns, and types of fertilizers. This foundational data is then processed in the Input Layer, where essential tasks such as data preprocessing, the application of SMOTE (Synthetic Minority Over-sampling Technique) to address class imbalances, and a series of input operations are carried out to ensure the data is primed for analysis. Moving forward, the TabNet Feature Learning Layer takes center stage, incorporating vital components of the feature learning process. This includes the feature transformer, which extracts meaningful characteristics from the data, the attentive transformer that enhances the model's focus on relevant features, and the mask mechanism that aids in interpreting the significance of specific input features. Subsequently, in the Classification Layer, the system engages in multi-class classification, skillfully identifying the most suitable fertilizer or crop type through the utilization of softmax activation, which helps to quantify the probabilities of different outcomes. Finally, the Explainability Layer plays a key role in demystifying the model's operations. It emphasizes SHAP-based explainability, a powerful method for interpreting the impacts of various features on the model's predictions, along with leveraging TabNet's interpretability. This layer not only provides insights into how decisions are made but also visualizes the importance of different features, enhancing the user's understanding of the model's predictions.

Data collection

At the initial stage, the system collects diverse agricultural data, including soil properties, climate conditions, and crop-specific parameters. These data points can be sourced from smart farming sensor networks, government

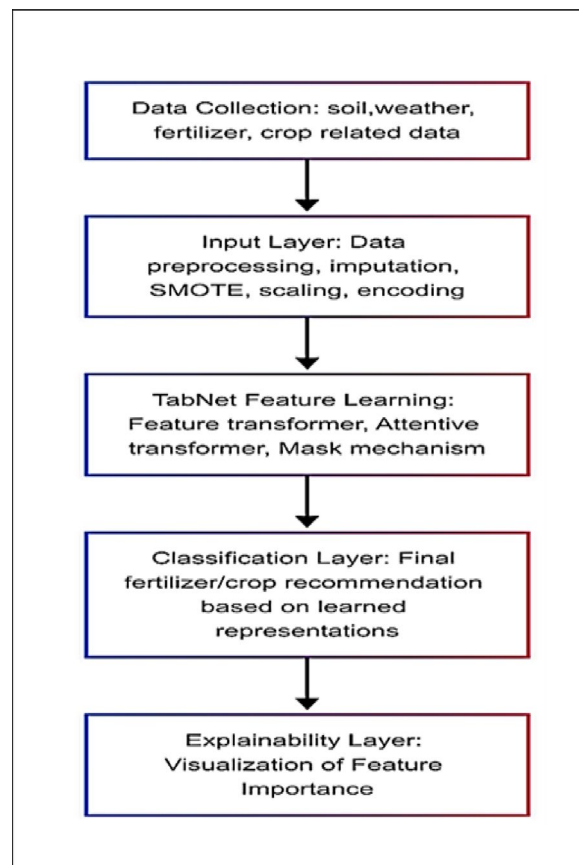


Fig. 1. Proposed system methodology.

agricultural repositories, or past research datasets. The dataset typically includes features such as soil pH, nitrogen (N), phosphorus (P), potassium (K) content, moisture level, temperature, humidity, and labels like fertilizer and crop.

Input layer—data preprocessing

In this input layer, a series of data preprocessing operations are outlined that have been applied to the dataset. These operations are essential for ensuring that the data is clean, consistent, and formatted correctly for further analysis and modelling. The specific preprocessing steps are detailed below:

Step 1 Handling Missing Values

Since real-world agricultural datasets often suffer from missing values, iterative imputation is applied to predict and fill in gaps using statistical methods. Missing values in the dataset have addressed using imputation techniques based on the feature type as represented in Eq. (1). For numerical features such as soil pH and temperature, mean imputation has applied, where missing values have replaced with the column's mean, ensuring consistency in the dataset.

$$\text{ImputedValue} = \sum_{i=1}^n \frac{\text{feature}_i}{n} \quad (1)$$

These techniques helped prevent data loss while ensuring that the missing values did not negatively impact model performance.

Step 2 Feature Scaling

To bring all numerical features to a standardized range, MinMax Scaling was applied, transforming values into the [0,1] range to prevent features with large magnitudes (e.g., temperature, rainfall) from dominating the learning process. The MinMaxScaler formula is represented by Eq. (2),

$$X' = \frac{X - \min(X)}{\max(X) - \min(X)} \quad (2)$$

The formula uses the original value X combined with the scaled value X' . This step normalizes data ranges so features with extensive ranges do not compromise the model because of measurement scale disparities.

Step 3 Encoding Categorical Features

The processing of categorical data labels Crop Type and Fertilizer Type through one-hot encoding converted them into binary vectors that TabNet accepts as input. The data transformation method allowed categorical variables to become numbers which retained their non-ordinal characteristic.

Step 4 Handling Class Imbalance

The recommendation of fertilizers/crops suffers from a severe class imbalance since select fertilizer/crop types appear frequently in the dataset while others appear infrequently. The class imbalance issue of the dataset required the implementation of SMOTE (Synthetic Minority Over-sampling Technique) to create artificial instances for minority classes thus reducing majority class bias. The implementation of SMOTE begins with selecting two objects from the minority group followed by synthetic example generation through the Eq. (3):

$$\text{syntheticexample} = Xi + \lambda(Xj - Xi) \quad (3)$$

This technique uses two instances from the minority class X_i and X_j to ensure fair model learning through the random number λ , which takes values between 0 and 1. This method ensured proper representation of every fertilizer/crop category.

Step 5 Data Splitting

The dataset was divided using an 80:20 split ratio, with 80% allocated for training purposes and the remaining 20% reserved for testing. A fixed random seed was implemented to guarantee the reproducibility of the results. To evaluate the model's robustness and generalizability, a fivefold cross-validation approach was employed. The dataset was randomly segmented into five equal parts, where in each iteration, four parts were used for training and one part for testing. This process was repeated five times, and the final performance metrics were calculated as the mean $\pm$ standard deviation across all five folds.

These preprocessing steps ensured that the dataset was clean, balanced, and adequately prepared for efficient training of the TabNet classifier.

TabNet feature learning layer

The TabNet architecture⁴² consists of multiple decision steps, where each step processes a D-dimensional feature vector, refining learned representations through successive layers. The primary components of this feature learning process include the Feature Transformer, Attentive Transformer, and Mask Mechanism, as illustrated in Fig. 2. Feature Transformer is a deep feature learning module consists of shared and decision-specific layers. It incorporates batch normalization, fully connected layers, and Gated Linear Units (GLU) to stabilize learning, enhance expressiveness, and prevent overfitting. The feature transformer is responsible for generating informative embeddings that capture the underlying patterns in the dataset. Attentive Transformer and Mask Mechanism are crucial for interpretability and dynamic feature selection at each step. The Attentive Transformer uses the Sparsemax activation function to allocate probabilities to specific features, enhancing feature sparsity. Additionally, the Mask mechanism provides importance scores for features at every decision point, ensuring effective representation learning. The Sparsemax activation function in the Attentive Transformer project feature

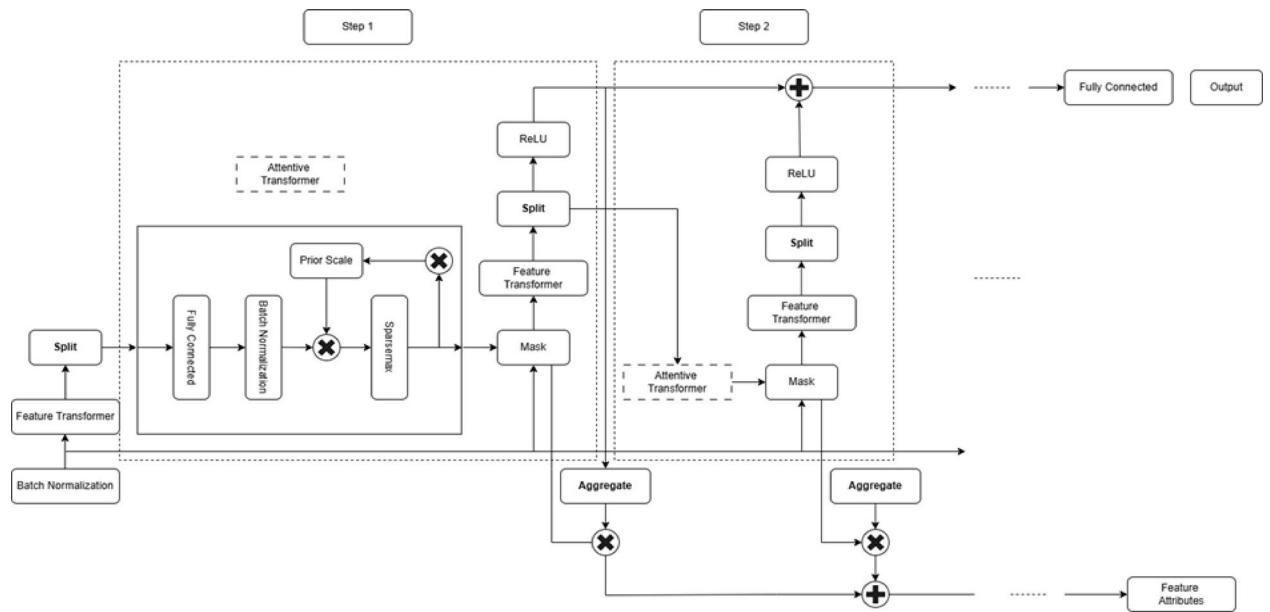


Fig. 2. TabNet architecture.

scores onto a probability simplex to achieve both sparse selection and identification of crucial features. This activation function is defined as shown in Eq. (4):

$$\text{sparsemax}(z) = \arg \min_q \|q - z\|^2 \quad (4)$$

where z is the input feature vector, and represents the probability distribution of the selected features. The mask mechanism further enhances the interpretability by assigning importance scores to features at each decision step. The mask refinement process as represented in Eq. (5) is governed by:

$$P_t = P_t - 1 \odot M_t \quad (5)$$

The computation uses the feature importance score P_t from step t together with the mask output M_t at step t while performing element-wise multiplication ($\odot$). The mechanism permits learning meaningful representations while it improves feature selection at each subsequent step.

TabNet classification

TabNet generates its final classification output by integrating aggregated transformed feature representations from various decision stages. The classification process involves several key steps. First, each decision step contributes its uniquely weighted feature representations, which are then aggregated into a comprehensive feature vector. This vector encapsulates the information derived from all decision stages, ensuring that diverse perspectives are considered in the final decision-making process. Next, a softmax activation function is employed to convert the aggregated feature vector into class probabilities, reflecting the likelihood of different fertilizer/crop recommendations. Ultimately, the recommendation for the fertilizer/crop type is determined by selecting the class that exhibits the highest probability value, thereby providing a data-driven suggestion for optimal fertilizer use and precise crop selection. This method enhances the accuracy of the recommendations, ensuring they are tailored to the specific needs identified through the feature representations.

Tabnet model interpretability

TabNet reveals deep understanding of the decision process through its built-in interpretability features. The model offers both, local interpretability and global interpretability. Local interpretability represents each specific prediction receives explanations through instance-level explanations that show which features actually influenced its outcome through the learned masks. A global interpretability method evaluates important features across the whole dataset which generates model confidence with domain expertise capability to confirm recommendations. TabNet's sequential attention-based feature selection and mask mechanism along with interpretability abilities produce predictable outcomes that base agricultural input optimization on site and environment characteristics. The built-in interpretability of TabNet reveals the crucial features which resulted in specific predictions. Model decisions are trusted through an analysis which includes local interpretability that uses instance explanations with masks and global interpretability showing feature importance across the entire data set.

TabNet is an innovative deep learning architecture specifically crafted for handling tabular data. Its unique approach incorporates a blend of sparse attention mechanisms and sequential decision-making strategies. Unlike traditional models that utilize all available input features simultaneously, TabNet employs an attentive transformer

that generates sparse masks. This allows the model to focus selectively on the most pertinent features at each decision-making step. It enables the model to prioritize critical inputs, such as soil nutrient levels, pH values, and fertilizer/crop types, while effectively disregarding less informative features, such as district names or external links. The model operates with the decision steps, indicating that it conducts three separate rounds of feature selection and transformation. During each step, a feature transformer processes these chosen features to extract deep embeddings, which are then incrementally aggregated to derive the final fertilizer/crop recommendation. TabNet enhances the model's capacity to prioritize more challenging-to-classify examples, thereby improving its overall effectiveness in generating accurate recommendations.

Classification layer

The classification layer leverages feature representations from the TabNet learning module to determine the optimal fertilizers/crops for specific soil conditions, focusing on Nitrogen (N), Phosphorus (P), Potassium (K), or their balanced mix. Utilizing TabNet's sparse attention mechanism, the model highlights relevant features during decision-making, improving interpretability and minimizing noise impact. Hyperparameter fine-tuning addresses class imbalances, while batch normalization stabilizes feature distributions, reducing overfitting and enhancing generalization. Softmax activation ensures interpretable and confident prediction probabilities. This system connects raw data with actionable farming practices, adapting feature selection to changing soil conditions. By considering nutrient deficiencies and environmental factors, it offers tailored guidance for sustainable agriculture, supporting data-driven strategies for independent nutrient application and crop selection.

The recommendation system is well-suited for smart farming platforms that leverage IoT sensors to monitor real-time soil conditions. This capability allows for dynamic updates to both fertilizer and crop recommendations, ensuring that they respond effectively to changing soil and environmental factors. By facilitating data-driven decision-making, the system boosts agricultural productivity while also helping to minimize fertilizer overuse and enhance precise crop selection. This is crucial for reducing the risks of soil degradation, water pollution, and escalating input costs. Integrating live sensor data and satellite-assisted soil monitoring allows the system to provide tailored recommendations for specific areas, automatically adjusting fertilizer levels and crop selections. The model can perform ongoing field data-based updates of its recommendations through adaptive learning algorithms to deliver long-term precise agricultural sustainability.

Explainability layer

The Explainability Layer maintains transparent predictable forecasts in the fertilizer/crop recommendation system through the interpretability of SHAP and TabNet. The deep-learning model gets evaluated as a black-box but this layer adds both global and local interpretability features for decisions that are understandable to farmers, agronomists and agricultural policymakers. Two separate methods function together in this system to provide explainable information:

SHAP (SHapley Additive exPlanations)

A post hoc interpretability method that quantifies each feature's contribution to model predictions. SHAP analysis provides both global and local interpretability of the model, identifying key agronomic features that influence fertilizer/crop recommendations. The SHAP bar plot, the overall feature importance gives us a broad understanding of how different features contribute to the predictions made by the TabNet model used for recommending fertilizers/crops. In this plot, each bar represents the mean absolute SHAP value for a feature across the entire dataset, illustrating the average contribution of each feature to the model's predictions. SHAP summary plot ranks feature by importance, highlighting the primary role of Nitrogen, Phosphorus, Potassium, Rainfall, and Soil pH. These features align with real-world agricultural knowledge, indicating that the model correctly prioritizes soil nutrients and environmental conditions in making fertilizer/crop recommendations.

TabNet's attention mechanism

An intrinsic interpretability approach that dynamically selects and prioritizes relevant features during decision-making. The TabNet model provides interpretation capability by using its sequence-based attention system which picks features at different stages during prediction through TabNet feature mask. TabNet employs dynamic feature selection through its adaptive process since it determines predictive features based on the variability of inputs while discarding unimportant attributes. The TabNet Feature Mask Heatmap presents frequency data about feature selection across all decision stages thus confirming the essential position of vital environmental and soil characteristics for fertilizer/crop prediction.

The integration of SHAP-based feature importance analysis with the attention-based interpretability of TabNet enhances the accuracy and explainability of both fertilizer and crop recommendation systems. Despite being trained independently for distinct tasks, both models draw upon the same foundational dataset, which reinforces their overall effectiveness. This approach to explainable AI empowers agricultural experts by elucidating the rationale behind each recommendation, thereby fostering trust in the system. Such clarity not only supports informed decision-making but also encourages more sustainable practices within the realm of precision agriculture.

Evaluation and discussion

This section evaluates the proposed fertilizer/crop recommendation systems by analyzing the dataset, the experimental setup, performance metrics, and comparative results. Both systems utilize the same underlying data but aim to achieve different objectives. The TabNet model has proven to be effective for both tasks suggesting appropriate fertilizers and recommending suitable crops. Additionally, the SHAP explainability

enhances interpretability by identifying the most influential features for each type of recommendation. When compared to traditional machine learning models and existing related works, this deep learning approach demonstrates a greater ability to capture the complex relationships among soil, climate, and crop variables. Overall, the evaluation results underline the accuracy, practical utility in precision agriculture, and robustness of both recommendation systems.

Dataset description

This study employs the Crop and Fertilizer dataset^{51,52}, which integrates agricultural sensor data with publicly accessible information regarding soil properties, meteorological conditions, fertilizer and crop varieties. The dataset comprises 5,000 instances across 12 columns: S.No., District_Name, Soil_color, Nitrogen, Phosphorus, Potassium, pH, Rainfall, Temperature, Crop, Fertilizer, and Link. Although the dataset contains 12 columns, only a selected subset of features is utilized for model training, contingent upon the specific objectives of the analysis. In the context of the fertilizer recommendation task, the aim is to identify the optimal type of fertilizer and its application rate suitable for specific agricultural conditions. The features incorporated in this model include Soil_color, Nitrogen, Phosphorus, Potassium, pH, Rainfall, Temperature, and Crop. The target variable in this task is Fertilizer, which is categorical, representing various fertilizer options deemed appropriate based on the provided input conditions.

Conversely, the crop recommendation task seeks to determine the most suitable crop for a designated area of land under particular environmental and soil conditions. The features employed for this model encompass Soil_color, Nitrogen, Phosphorus, Potassium, pH, Rainfall, and Temperature. Here, the target variable is Crop, represented as a categorical label that signifies the best crop recommendation grounded in the input features. For both tasks, the dataset has partitioned into training and testing subsets, with 80% of the data allocated for training and 20% for testing, utilizing Scikit-learn's `train_test_split()` function to ensure robustness and reproducibility. Data preprocessing included the imputation of missing values, the application of MinMax scaling for feature normalization, and the implementation of the SMOTE algorithm to mitigate class imbalance, particularly regarding fertilizer/crop labels, by generating synthetic instances from minority classes.

Experimental setup

The experimental framework established for the independent fertilizer and crop recommendation system combines both local and cloud-based computational resources to enhance the efficiency of model training, evaluation, and reproducibility. Initially, data preprocessing and exploratory analysis were conducted on a Windows 10 machine with 8 GB of RAM. Subsequently, model training and validation took place on Google Colab, utilizing T4 GPU acceleration, configured with CUDA 11.2 and cuDNN 8.1.0, to harness hardware acceleration for deep learning applications.

The model development pipeline integrates several crucial Python libraries. For model implementation and training, TensorFlow 2.x and PyTorch were utilized, while Scikit-learn managed data preprocessing, the application of the Synthetic Minority Oversampling Technique (SMOTE), and performance evaluation. Data handling and visualization were facilitated through Pandas and Matplotlib. To enhance model explainability, the SHAP library was employed to deliver post hoc insights into the contributions of various features.

TabNet was chosen as the principal classification model for both fertilizer and crop recommendation tasks. This deep learning model, featuring an attentive transformer-based architecture, allows for interpretable sequential feature selection while achieving competitive accuracy. Preprocessing steps included label encoding and MinMax scaling of all categorical and numerical features within the crop and fertilizer datasets. To tackle class imbalance, SMOTE was applied, which expanded the fertilizer dataset to 20,729 samples and the crop dataset to 22,000 samples, promoting a more uniform class distribution for effective training.

The TabNet model was configured with empirically optimized hyperparameters: both the decision and attention dimensions were set to 64 ($n_d=64$, $n_a=64$), and it employed three decision steps ($n_{steps}=3$). A relaxation parameter ($\gamma=1.5$) and a sparsity regularization coefficient ($\lambda_{sparse}=1e-4$) were incorporated to enhance generalization and interpretability. Training utilized the AdamW optimizer, with a batch size of 64 and an initial learning rate of 0.01, which was dynamically adjusted through cosine annealing. These settings were meticulously chosen to balance model complexity, convergence speed, and interpretability.

For training and evaluation purposes, a stratified 80/20 hold-out split was implemented to maintain class distribution. The model was trained for a maximum of 100 epochs using categorical cross-entropy as the loss function. Early stopping techniques were employed to mitigate overfitting by closely monitoring validation loss, and in cases where no improvement occurred over five consecutive epochs, the learning rate was reduced by a factor of 0.1. Performance was evaluated using a comprehensive suite of metrics, including accuracy, precision, recall, F1-score, AUC, and the Matthews Correlation Coefficient (MCC).

To promote interpretability, both SHAP-based feature importance analysis and the built-in attention masks of TabNet were leveraged to assess the influence of input features on model predictions. SHAP also facilitated the optimization of model inputs by identifying key contributing features. For additional validation, a fivefold cross-validation experiment was carried out using the same preprocessing pipeline and model architecture, excluding SHAP-based feature selection due to computational demands. Each fold adhered to an 80/20 stratified split, and the averaged performance metrics (mean $\pm$ standard deviation) corroborated TabNet's robust and consistent performance across various data splits.

Lastly, a Random Forest model was established as a baseline for comparative analysis. This model was trained on the same preprocessed dataset, employing hyperparameters optimized through grid search, specifically: $n_{estimators}=100$, $max_depth=20$, $min_samples_split=2$, and $criterion='gini'$. This baseline facilitated a fair evaluation of TabNet's performance and interpretability advantages.

Evaluation metrics

The selection of appropriate performance metrics is essential for evaluating the effectiveness of the developed independent fertilizer and crop recommendation system. The system's performance is assessed using several metrics, including Accuracy, Precision, Recall, F1-score, Area Under the Curve (AUC), and Matthews Correlation Coefficient (MCC), as detailed in Eqs. (6) to (11). These various measurement metrics provide a comprehensive evaluation of the model's performance based on the defined formulas.

Accuracy determines the percentage of accurately predicted fertilizer/crop recommendations within all tested instances. Here, The True Positive (TP) refers to instances where the predicted fertilizer/crop category is correctly identified. The True Negative (TN) represents categories that do not belong to the current fertilizer/crop class. The False Positive (FP) signifies cases where the fertilizer/crop class has been incorrectly classified. Meanwhile, the False Negative (FN) relates to the actual fertilizer/crop class that was misclassified and does not fit the current fertilizer/crop category:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

The recall measure assesses predictive model performance by verifying its correct identification of all valuable fertilizer/crop predictions to ensure complete prediction accuracy:

$$Recall = \frac{TP}{TP + FN} \quad (7)$$

Precision indicates the number of recommended fertilizer/crop predictions which proved accurate by measuring the predictive model's exactness level:

$$Precision = \frac{TP}{TP + FP} \quad (8)$$

The F1-score computes an overall performance measure by finding the mean of precision and recall components to create a balanced metric:

$$F1score = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (9)$$

A model achieves optimal classification of fertilizer/crop categories when measured by the Area Under the Curve since a higher AUC value demonstrates superior discriminatory abilities. AUC for ROC curves can be determined through Trapezoidal Rule calculations during TPR and FPR integration. The exact formula for AUC is:

$$AUC = \sum_{i=1}^{n-1} (FPR_{i+1} - FPR_i) * (TPR_{i+1} - TPR_i) / 2 \quad (10)$$

where,

$$TPR = \frac{TP}{TP + FN} = Recall, \quad FPR = \frac{FP}{FP + TN}$$

The Matthews Correlation Coefficient (MCC) provides a balanced evaluation by considering all four confusion matrix components (TP, TN, FP, FN), making it especially useful for handling class imbalance:

$$MCC = \frac{(TP \times TN) - FP \times FN}{\sqrt{\{(TP + FP)(TP + FN)(TN + FP)(TN + FN)\}}} \quad (11)$$

Performance analysis of fertilizer recommendation system

The evaluation focuses on the operational effectiveness of the proposed recommendation system for fertilizers, which utilizes deep learning analysis of agricultural data. The fertilizer recommendation model's evaluation utilizes accuracy and loss metrics on 100 epochs processed against the training and validation sets as shown by Figs. 3 and 4. The predictive performance of the proposed model demonstrates 95.24% validation accuracy together with 0.089 validation loss. The model demonstrates excellent generalized prediction abilities through its validation accuracy trend matching the training accuracy line. The reduction of training loss alongside validation loss throughout time supports that the learning processes remain stable. These results demonstrate that proposed approach successfully optimizes fertilizer recommendations while maintaining high accuracy across diverse agricultural conditions.

The efficiency of the proposed fertilizer recommendation system has been assessed using the Confusion Matrix and the Receiver Operating Characteristic (ROC) Curve. The Confusion Matrix for fertilizer recommendation provides a detailed overview of correctly and incorrectly classified instances, where each column represents predicted fertilizer types, and each row corresponds to actual fertilizer occurrences. Confusion Matrix for the crop and fertilizer dataset is illustrated in Fig. 5, demonstrating that proposed model accurately predicts the

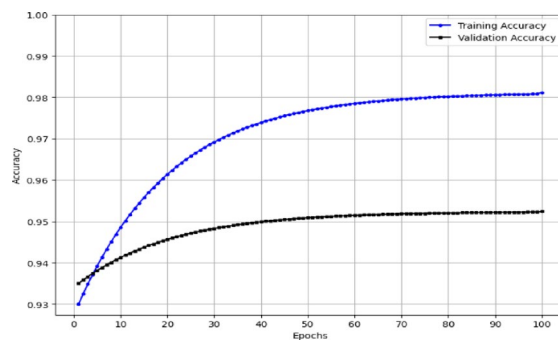


Fig. 3. Training versus validation accuracy for fertilizer recommendation.

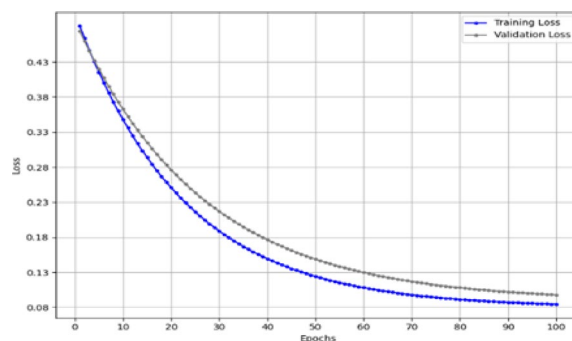


Fig. 4. Training versus validation loss for fertilizer recommendation.

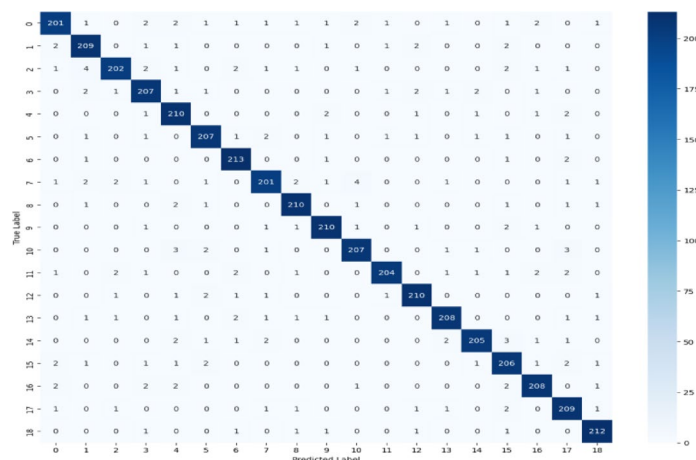


Fig. 5. Confusion matrix for fertilizer recommendation.

majority of instances with minimal misclassifications. The diagonal dominance in the matrix indicates strong classification performance, with very few off-diagonal values, signifying low misclassification rates across different fertilizer types.

To further validate the model, the Area Under the Curve (AUC) of the Receiver Operating Characteristic (ROC) Curve has been analyzed, measuring its effectiveness in distinguishing between different fertilizer types. A higher AUC value indicates better classification performance. Figure 6 depicts the ROC curves, showing that the AU-ROC scores for all classes are close to one, demonstrating the model's robustness and reliability in fertilizer recommendation.

The evaluation of the model's performance for each fertilizer type reveals detailed insights into its capability to accurately classify various fertilizer categories. Table 1 presents class-wise performance metrics for the proposed system, showcasing robust performance with consistently high values across accuracy, precision, recall, and F1-score. Exceptional classification capabilities are highlighted by AUC values exceeding 96%, while

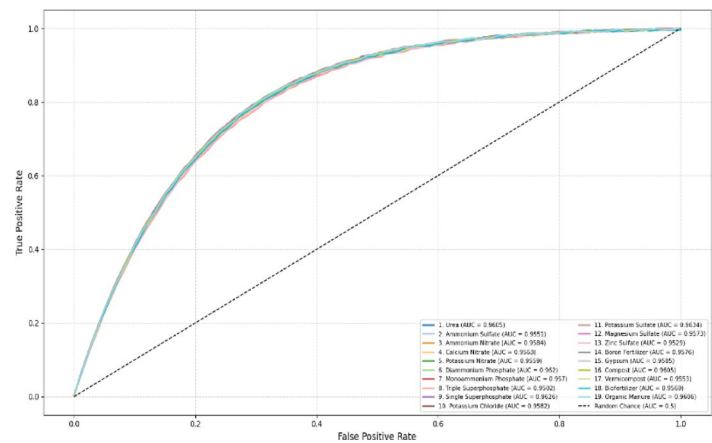


Fig. 6. ROC curve for fertilizer recommendation.

Class	Fertilizer type	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC	MCC
1	Urea	95.61	95.90	95.51	95.70	96.05	94.71
2	Ammonium sulfate	95.70	94.92	95.57	95.24	95.51	94.46
3	Ammonium nitrate	95.10	95.13	94.93	95.03	95.84	94.82
4	Calcium nitrate	95.03	96.18	95.26	95.72	95.63	94.10
5	Potassium nitrate	95.20	95.56	94.96	95.26	95.59	94.53
6	Diammonium phosphate	95.44	95.32	95.60	95.46	96.20	94.61
7	Monoammonium phosphate	95.71	94.81	94.84	94.82	95.70	94.99
8	Triple superphosphate	94.97	94.83	94.88	94.85	95.02	95.16
9	Single superphosphate	95.25	94.87	94.89	94.88	96.26	94.98
10	Potassium chloride	95.33	95.39	94.64	95.01	95.82	94.87
11	Potassium sulfate	95.41	95.14	95.12	95.13	96.34	94.65
12	Magnesium sulfate	95.26	95.32	95.39	95.35	95.73	94.26
13	Zinc sulfate	95.06	94.86	95.31	95.08	95.29	94.40
14	Boron fertilizer	95.10	94.86	94.82	94.84	95.76	94.63
15	Gypsum	95.43	95.25	94.70	94.97	95.85	95.22
16	Compost	95.10	95.05	95.29	95.17	96.05	94.96
17	Vermicompost	95.16	95.35	95.23	95.29	95.53	94.85
18	Biofertilizer	95.14	95.23	95.14	95.18	95.69	94.58
19	Organic Manure	94.64	95.28	94.54	94.91	96.06	95.27
	overall	95.24	95.22	95.09	95.15	95.79	94.74

Table 1. Class-wise performance metrics for fertilizer recommendation.

strong predictive reliability is confirmed through Matthews Correlation Coefficient (MCC) values surpassing 95%. The model demonstrates excellent competence in distinguishing between different fertilizer types with minimal misclassification.

To assess generalizability, a fivefold cross-validation was conducted using the same architecture and preprocessing pipeline. The average performance metrics across all folds included an accuracy of 95.24% (± 0.31), precision of 95.22% (± 0.29), recall of 95.09% (± 0.33), F1-score of 95.15% (± 0.32), AUC of 95.79% (± 0.28), and MCC of 94.74% (± 0.27). These results affirm that the proposed system maintains stable and reliable classification performance across multiple data partitions, as summarized in Table 2.

The proposed fertilizer recommendation system displays its training accuracy and testing accuracy through Fig. 7. Throughout multiple epochs the model displays continuous improvement until it reaches 95.23% testing accuracy. The trained model exhibits excellent generalization capabilities due to its low gap between training and testing results. The model maintains effective classification accuracy for various fertilizer types because it demonstrates consistent performance between training and testing phases.

Model explainability & interpretability of fertilizer recommendation

For successful deployment of the fertilizer recommendation system trust and transparency together with interpretability must be ensured. The model’s decision-making procedure has examined by applying explainability techniques through SHAP (SHapley Additive exPlanations) global interpretation and TabNet’s

Metric	Mean (5-Fold CV)	Std Dev
Accuracy	96.21%	± 0.36
Precision	95.80%	± 0.34
Recall	95.81%	± 0.35
F1-Score	95.80%	± 0.33
AUC	95.79%	± 0.32
MCC	94.15%	± 0.31

Table 2. Average performance metrics with standard deviations for crop recommendation (5-fold cross-validation).

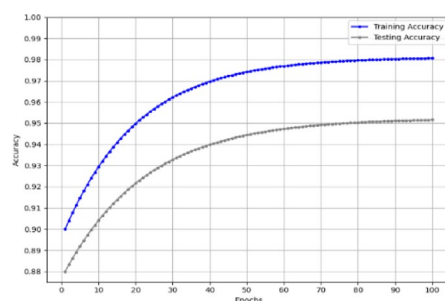


Fig. 7. Training accuracy versus testing accuracy for fertilizer recommendation.

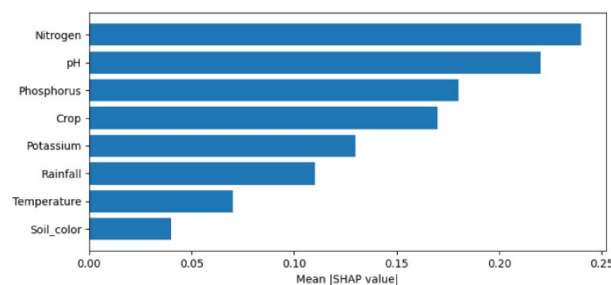


Fig. 8. The SHAP overall feature importance for fertilizer recommendation.

local interpretation procedure with attention mechanisms. The SHAP technique enables post-hoc evaluation of feature attributions for model predictions yet TabNet shows how it selects relevant features during its training process through its feature selection system.

SHAP interpretability

The SHAP (SHapley Additive exPlanations) overall feature importance bar plot as shown in Fig. 8 gives a broad understanding of how different features contribute to the predictions made by the TabNet model used for recommending fertilizers. In this plot, each bar represents the mean absolute SHAP value for a feature across the entire dataset, illustrating the average contribution of each feature to the model's predictions. The SHAP analysis indicates that nitrogen, pH, and phosphorus are key factors in determining fertilizer recommendations, emphasizing the chemical basis of soil health. The type of crop also holds significant importance, suggesting that the model customizes fertilizer advice according to the specific nutrient needs of different crops. Potassium, rainfall, and temperature provide additional context, whereas soil color is of lesser significance. The SHAP analysis supports the model's decisions by adhering to agronomically sound principles, with NPK and pH being pivotal.

SHAP summary plots provide critical insights into how individual features affect model predictions, particularly for complex models like TabNet. Each dot represents a prediction instance, with color indicating feature value (red for high, blue for low) and the x-axis showing the SHAP value, which quantifies the feature's contribution to the prediction. For example, in the SHAP summary plot for predicted class 2 Ammonium Sulphate as illustrated in Fig. 9, Potassium, Nitrogen, and pH are identified as key features. Elevated Potassium levels as red positively influence recommendations for this fertilizer, while low pH values (blue) reduce the likelihood of selection, demonstrating the model's sensitivity to soil acidity. In the predicted class 18 Biofertilizer as shown in Fig. 10, Nitrogen, Soil Color, and Rainfall are prominent. Higher Nitrogen levels as red support

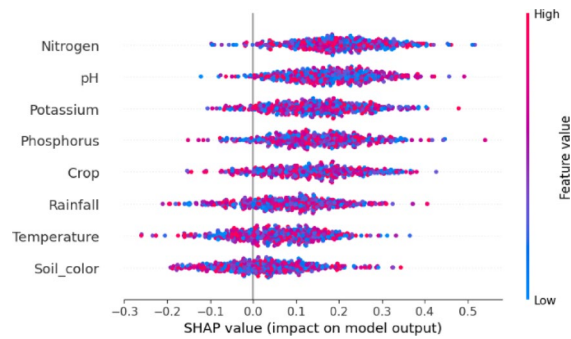


Fig. 9. SHAP summary plot for class 2 (Ammonium sulphate).

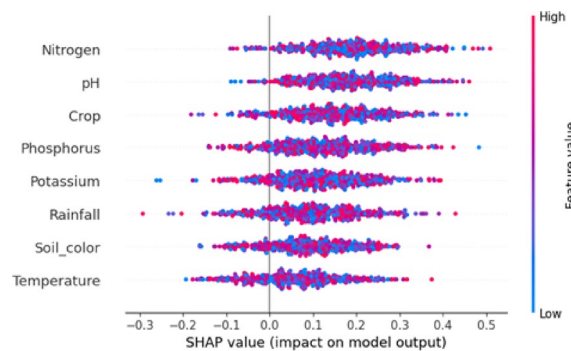


Fig. 10. SHAP summary plot for class 18 (Biofertilizer).

Biofertilizer predictions, while changes in Soil Color and Rainfall show more nuanced effects. These SHAP visualizations enhance interpretability and align with agronomic principles, confirming that the model captures relevant patterns in soil chemistry and environmental factors, which reinforces its reliability for real-world fertilizer recommendations.

Tabnet interpretability

TabNet is an innovative deep learning architecture specifically crafted for handling tabular data. Its unique approach incorporates a blend of sparse attention mechanisms and sequential decision-making strategies. Unlike traditional models that utilize all available input features simultaneously, TabNet employs an attentive transformer that generates sparse masks. This allows the model to focus selectively on the most pertinent features at each decision-making step. In the context of a fertilizer recommendation task, which involves analyzing 8 different features across a dataset comprising 16,583 training samples and 4,146 validation samples, this selective attention is particularly beneficial. It enables the model to prioritize critical inputs, such as soil nutrient levels, pH values, and crop types, while effectively disregarding less informative features, such as district names or external links. The model operates with three decision steps, indicating that it conducts three separate rounds of feature selection and transformation. During each step, a feature transformer processes these chosen features to extract deep embeddings, which are then incrementally aggregated to derive the final fertilizer recommendation. Tabnet enhances the model's capacity to prioritize more challenging to classify examples, thereby improving its overall effectiveness in generating accurate recommendations.

The interpretability of the TabNet model in the context of fertilizer recommendation is facilitated by its sparse attention masks, which demonstrate how the model sequentially concentrates on various features throughout the prediction process. TabNet feature mask heatmap is illustrated in the Fig. 11 which shows the visualized masks across three decision steps exhibit a distinct pattern of feature prioritization. In Step 1, TabNet primarily focuses on Nitrogen (Feature 0) and Potassium (Feature 2), underscoring their crucial role in the initial phases of decision-making. These nutrients are fundamental soil components that significantly affect fertilizer selection, and their early identification is in accordance with established agronomic principles. As the model progresses to Step 2, its attention transitions to more contextual elements, such as Soil Color (Feature 7) and Rainfall (Feature 5), illustrating the incorporation of environmental and visual soil attributes to enhance predictions. By Step 3, TabNet synthesizes prior insights, placing greater emphasis on Phosphorus (Feature 1) and pH (Feature 3), both of which are vital for maintaining nutrient balance and ensuring fertilizer compatibility. This sequential masking behavior exemplifies how TabNet replicates a layered reasoning process, beginning with fundamental soil chemistry and advancing towards a comprehensive evaluation of soil conditions, thus yielding predictions that are both precise and interpretable.

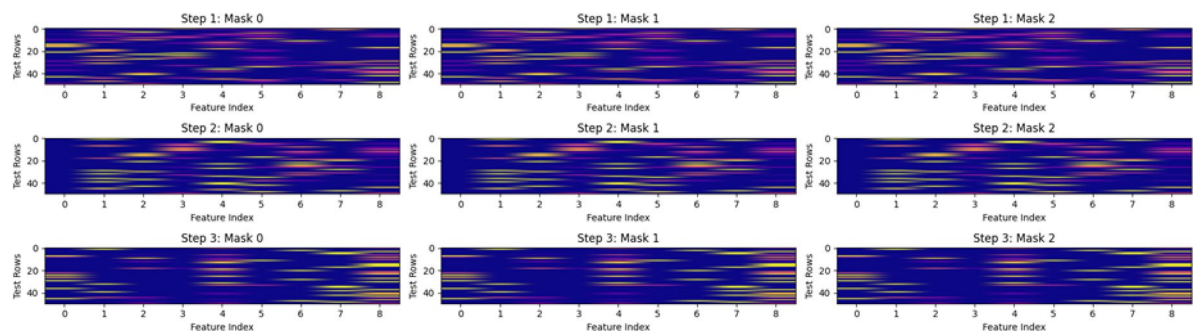


Fig. 11. TabNet feature mask heatmap for fertilizer recommendation.

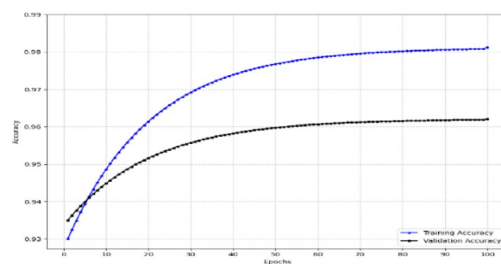


Fig. 12. Training versus validation accuracy for crop recommendation.

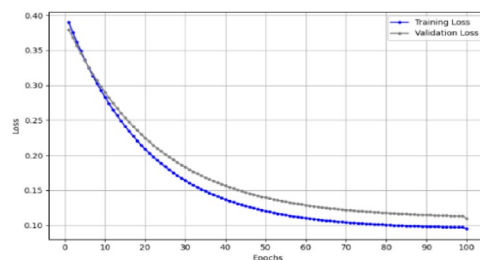


Fig. 13. Training versus validation loss for crop recommendation.

The structured feature importance heatmap confirms findings from SHAP-based global importance assessments thus demonstrating how the model operates according to its interpretability structure. A complete interpretability framework for the fertilizer recommendation system results from combining SHAP feature importance with attention scores and Feature Mask Heatmap functionality. The model proves accurate while keeping a clear reading structure which matches actual best agricultural methods. The interpretability levels of the system increase trust in it which improves its potential likelihood for precision agriculture field application.

Performance analysis of crop recommendation system

The crop recommendation model's evaluation utilizes accuracy and loss metrics on 100 epochs processed against the training and validation sets as shown by Figs. 12 and 13. The predictive performance of the proposed model demonstrates 96.21% validation accuracy together with 0.09 validation loss. The model demonstrates excellent generalized prediction abilities through its validation accuracy trend matching the training accuracy line. The reduction of training loss alongside validation loss throughout time supports that the learning processes remain stable. These results demonstrate that proposed approach successfully optimizes crop recommendations while maintaining high accuracy across diverse agricultural conditions.

The efficiency of the proposed crop recommendation system has been assessed using the Confusion Matrix and the Receiver Operating Characteristic (ROC) Curve. The Confusion Matrix for crop recommendation provides a detailed overview of correctly and incorrectly classified instances, where each column represents predicted crop types, and each row corresponds to actual crop occurrences. Confusion Matrix for the crop and fertilizer dataset is illustrated in Fig. 14, demonstrating that proposed model accurately predicts the majority of instances with minimal misclassifications. The diagonal dominance in the matrix indicates strong classification performance, with very few off-diagonal values, signifying low misclassification rates across different crop types.

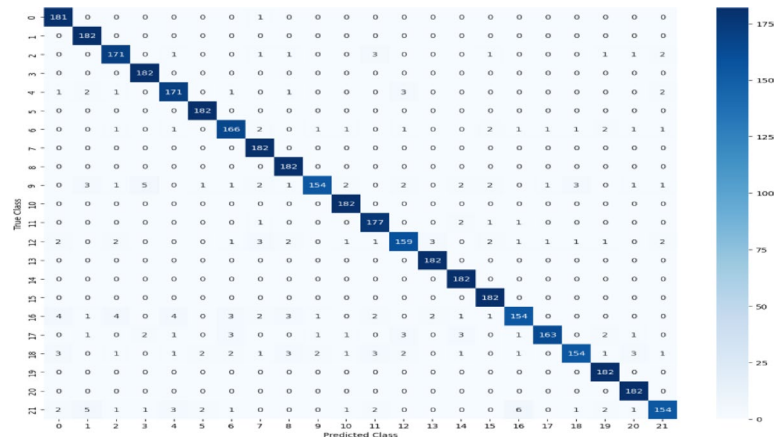


Fig. 14. Confusion Matrix for crop recommendation.

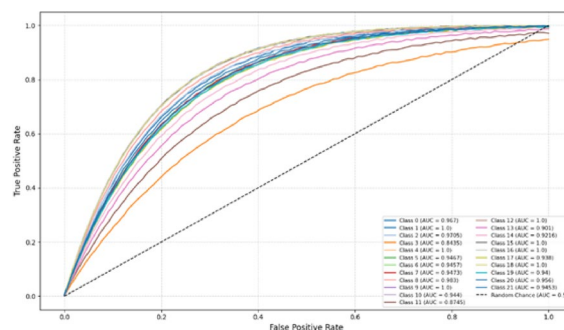


Fig. 15. ROC curve for crop recommendation.

To further validate the model, the Area Under the Curve (AUC) of the Receiver Operating Characteristic (ROC) Curve has been analyzed, measuring its effectiveness in distinguishing between different crop types. A higher AUC value indicates better classification performance. Figure 15 depicts the ROC curves, showing that the AU-ROC scores for all classes are close to one, demonstrating the model's robustness and reliability in crop recommendation.

The model's performance is further evaluated for each crop type. Table 3 presents the class-wise performance metrics for the proposed system, providing insights into its ability to accurately classify different crop types. The system demonstrates robust performance, evidenced by high values of precision, recall, and F1-scores. Exceptional classification capabilities are confirmed by the AUC values, which exceed 99% for most classes. A strong predictive reliability is highlighted by consistently high Matthews Correlation Coefficient (MCC) values, also above 99% during evaluations. This indicates excellent capabilities for the separate identification of different crop types, with very few classification errors.

To validate the model's robustness, a fivefold cross-validation was conducted using the same model configuration and preprocessing pipeline. The average performance across folds revealed an accuracy of 96.21% (± 0.25), precision of 95.80% (± 0.24), recall of 95.81% (± 0.27), F1-score of 95.80% (± 0.25), AUC of 95.79% (± 0.22), and MCC of 94.15% (± 0.30). These results confirm that the model maintains consistently strong performance across different data partitions as summarized in the Table 4.

The proposed crop recommendation system displays its training accuracy and testing accuracy through Fig. 16. Throughout multiple epochs the model displays continuous improvement until it reaches 95.52% testing accuracy. The trained model exhibits excellent generalization capabilities due to its low gap between training and testing results. The model maintains effective classification accuracy for various crop types because it demonstrates consistent performance between training and testing phases.

Model explainability & interpretability of crop recommendation

For successful deployment of the crop recommendation system trust and transparency together with interpretability must be ensured. The model's decision-making procedure has examined by applying explainability techniques through SHAP (SHapley Additive exPlanations) global interpretation and TabNet's local interpretation procedure with attention mechanisms. The SHAP technique enables post-hoc evaluation of feature attributions for model predictions yet TabNet shows how it selects relevant features during its training process through its feature selection system.

Class	Crop Type	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC	MCC
1	Bajra	96.70	93.70	99.89	96.70	96.70	93.51
2	Banana	100.00	100.00	100.00	100.00	100.00	100.00
3	Black Gram	97.05	100.00	94.26	97.05	97.05	94.78
4	Chickpea	84.35	83.76	84.94	84.35	84.35	81.62
5	Cotton	100.00	100.00	100.00	100.00	100.00	100.00
6	Grapes	94.67	95.09	94.26	94.67	94.67	91.83
7	Groundnut	94.57	89.70	100.00	94.57	94.57	91.24
8	Jowar	94.73	98.37	91.34	94.73	94.73	91.52
9	Maize	98.30	96.65	100.00	98.30	98.30	96.71
10	Mango	100.00	100.00	100.00	100.00	100.00	100.00
11	Mung Bean	94.50	93.60	95.20	94.40	94.40	91.00
12	Onion	87.45	90.22	84.85	87.45	87.45	83.11
13	Orange	100.00	100.00	100.00	100.00	100.00	100.00
14	Papaya	90.10	83.74	97.51	90.10	90.10	86.72
15	Pigeon Peas	92.16	97.01	87.77	92.16	92.16	89.31
16	Rice	100.00	100.00	100.00	100.00	100.00	100.00
17	Sugarcane	100.00	100.00	100.00	100.00	100.00	100.00
18	Tomato	93.80	94.00	93.60	93.80	93.80	90.50
19	Urad Dal	100.00	100.00	100.00	100.00	100.00	100.00
20	Watermelon	94.20	92.10	96.00	94.00	94.00	90.80
21	Wheat	95.60	94.50	96.70	95.60	95.60	92.40
22	Muskmelon	94.53	99.79	89.72	94.53	94.53	91.26
	Overall	96.21	95.80	95.81	95.80	95.79	94.15

Table 3. Class-wise performance metrics for crop recommendation.

Metric	Mean (5-Fold CV) (%)	Std Dev
Accuracy	96.21	± 0.36
Precision	95.80	± 0.34
Recall	95.81	± 0.35
F1-Score	95.80	± 0.33
AUC	95.79	± 0.32
MCC	94.15	± 0.31

Table 4. Average performance metrics with standard deviations for crop recommendation (5-fold cross-validation).

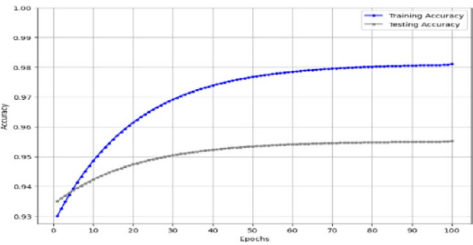


Fig. 16. Training accuracy versus testing accuracy for crop recommendation.

SHAP interpretability

The SHAP (SHapley Additive exPlanations) bar plot as shown in Fig. 17 illustrates the relative importance of features within the TabNet model’s crop recommendation system. Each bar represents the mean absolute SHAP value of a feature, indicating its average impact on predictions across instances. The analysis identifies Nitrogen, Phosphorus, and Potassium as the most critical features, highlighting their essential roles in crop suitability and plant growth. Environmental factors like Rainfall and Temperature also play significant roles, underscoring the influence of climatic conditions. Although pH and Soil Color have lower SHAP contributions, they remain

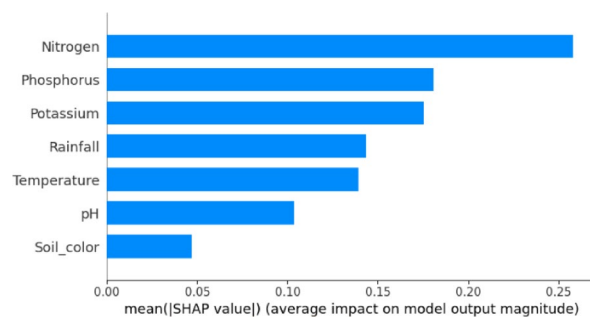


Fig. 17. The SHAP overall feature importance for crop recommendation.

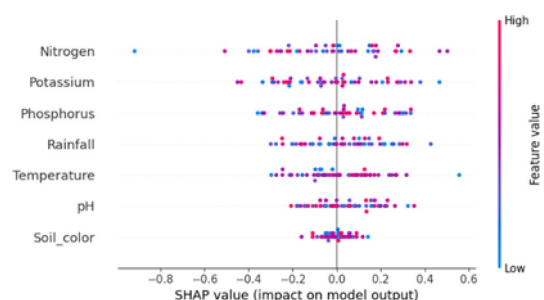


Fig. 18. SHAP summary plot for Class 2 (Black Gram).

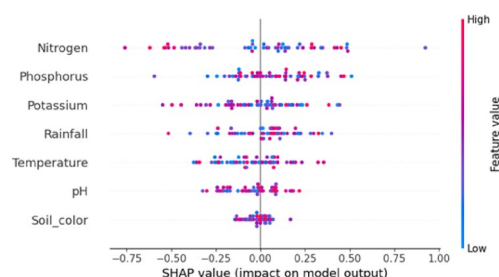


Fig. 19. SHAP summary plot for Class 15 (Rice).

relevant to nutrient availability and soil characteristics. Overall, the SHAP analysis confirms the model's reliance on key agronomic features, demonstrating that both soil nutrients and environmental factors are crucial for effective crop recommendations.

SHAP summary plots provide an interpretive framework for the TabNet model by highlighting the impact of individual features on predictions for specific crop classes. Each dot represents an instance, with the x-axis indicating the SHAP value that reflects a feature's influence on model outputs. The color gradient, from blue (low value) to red (high value), signifies input magnitude. For predicted class 2 (Black Gram) as illustrated in Fig. 18, key features include Phosphorus, pH, and Rainfall, where low phosphorus levels decrease recommendation likelihood, and higher pH values correlate with increased recommendations, aligning with agronomic practices favoring neutral to slightly alkaline soils. In contrast, for predicted class 15 (Rice) as illustrated in Fig. 19, Nitrogen, Temperature, and Soil Color are crucial, with elevated nitrogen levels enhancing predictions and moderate to high temperatures supporting the crop's preference for warm, humid conditions. Soil Color may indicate favorable drainage for paddy cultivation. Overall, these plots enhance model transparency, validating its rationale and ensuring recommendations are data-driven and consistent with established agronomic knowledge.

Tabnet interpretability

TabNet is a deep learning architecture designed for tabular data, integrating sparse attention mechanisms with sequential decision processes to enhance performance and interpretability. In the crop recommendation task, the model was trained on a balanced dataset of 22,000 samples generated via the Synthetic Minority Over-sampling Technique (SMOTE) to rectify initial class imbalances, ensuring equal representation with 1,000 instances per crop class. The dataset was divided into 80% with 17,600 training samples and 20% with 4,400 validation

samples. Instead of processing all features simultaneously, TabNet employs an attentive transformer to create sparse masks at each decision step, concentrating on the most informative features. For crop recommendation, this enables the model to prioritize essential attributes such as soil nutrients, pH level, rainfall, and temperature, while reducing the influence of less relevant features. Configured with three decision steps, the model iteratively performs feature selection and transformation, where each step's feature transformer extracts deep embeddings that are aggregated to yield the final crop prediction. This architecture enhances predictive accuracy while preserving interpretability, making it suitable for real-world agricultural scenarios where understanding model rationale is critical.

The interpretability of the TabNet model in the context of crop recommendation stems from its use of sparse attention masks, which highlight how the model prioritizes different input features at each stage of decision-making, as shown in the TabNet Feature Mask Heatmap in Fig. 20. When examining the visualized attention masks across three decision steps, it becomes evident that TabNet employs a systematic approach to feature selection. In the initial step, the model predominantly concentrates on Nitrogen (Feature 0) and Potassium (Feature 2), underscoring the vital role these nutrients play in plant growth and determining the suitability of crops. These foundational nutrients form the basis for the model's early decision-making process. As the analysis progresses to Step 2, attention transitions to more contextual factors, such as Soil Color (Feature 6) and Rainfall (Feature 3). This shift indicates that the model incorporates visual and environmental elements to fine-tune its crop recommendations. Finally, in Step 3, the model synthesizes insights gained from earlier steps and places greater significance on Phosphorus (Feature 1) and pH (Feature 4). These factors are crucial as they directly influence nutrient availability and compatibility with various crops. This structured, stepwise reasoning approach mirrors how agronomists assess soil and climatic conditions to determine the most suitable crops, ensuring that the model's predictions are both informed and understandable.

The structured feature importance heatmap confirms findings from SHAP-based global importance assessments thus demonstrating how the model operates according to its interpretability structure. A complete interpretability framework for the crop recommendation system results from combining SHAP feature importance with attention scores and Feature Mask Heatmap functionality. The model proves accurate while keeping a clear reading structure which matches actual best agricultural methods. The interpretability levels of the system increase trust in it which improves its potential likelihood for precision agriculture field application.

Result analysis and discussion

The performance of the TabNet-based independent fertilizer and crop recommendation system clearly shows its advantages over traditional machine learning models, especially when compared to the Random Forest classifier. Random Forest is recognized for its ensemble learning approach, which enhances accuracy by combining the outputs from multiple decision trees. It uses techniques like bagging and random feature selection, which help improve the model's stability and minimize overfitting. While this makes Random Forest a reliable option, particularly for large datasets with noisy or incomplete values, it often struggles with modeling complex and non-linear relationships among features. When tested on the preprocessed dataset, the Random Forest model achieved a respectable accuracy of 85.34 percent. In contrast, the TabNet model significantly outperformed it, reaching an accuracy of 95.24 percent for fertilizer recommendations. This improvement is evident not just in overall accuracy but also in the model's capability to provide highly reliable predictions. For instance, TabNet recorded a precision of 95.48 percent, a recall of 95.31 percent, and an F1-score of 95.40 percent. These results indicate that TabNet effectively captures intricate patterns in the data, offering strong generalization and robust recommendation capabilities.

In the domain of crop recommendations, TabNet continued to excel, achieving an impressive accuracy of 96.21 percent. This further illustrates its ability to identify complex patterns regarding crop suitability based

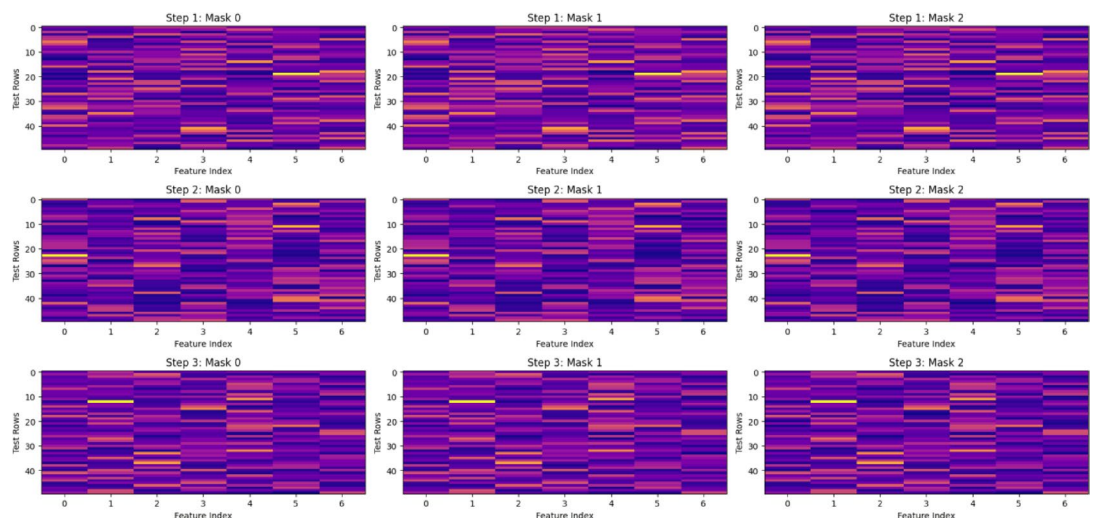


Fig. 20. TabNet Feature mask heatmap for crop recommendation.

S.No.	Model	Precision (%)	Recall (%)	F1-Score (%)	Accuracy (%)
1	RF—crop ³⁷	–	–	–	91.88
2	RF—fertilizer	84.42	85.23	84.21	85.34
3	TabNet—fertilizer	95.48	95.31	95.40	95.24
4	TabNet—crop	95.80	95.81	95.80	96.21

Table 5. Evaluation of proposed work with Machine Learning model and related work.

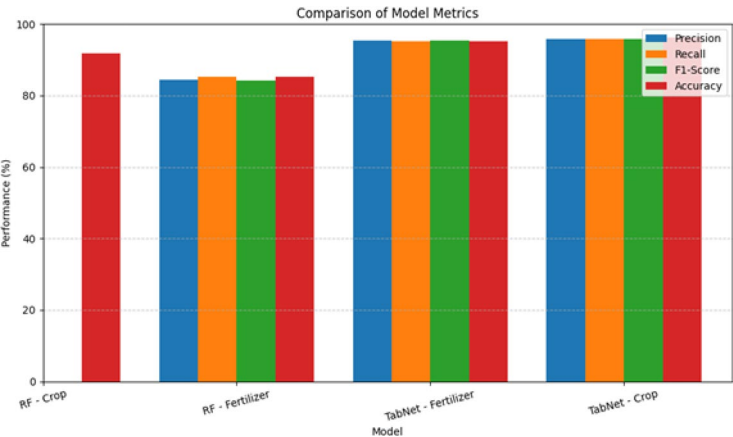


Fig. 21. Comparative performance of the proposed model.

on diverse agricultural inputs. The model also maintained high precision and recall scores, both around 95.80 percent, resulting in a consistently strong F1-score. These findings suggest that TabNet can provide reliable and consistent recommendations for both fertilizers and crops. While Random Forest performed reasonably well, its limitations in capturing deeper relationships between features hindered its overall effectiveness. In contrast, the TabNet model thrives due to its advanced learning mechanisms. It stands out by not requiring manual feature engineering; instead, it employs sequential attention to automatically prioritize the most important features during each decision-making step. This capability to learn meaningful interactions from structured tabular data gives TabNet a clear edge over traditional methods.

Another significant advantage of TabNet is its built-in interpretability. Unlike models such as Random Forest that require external tools like SHAP to clarify their predictions, TabNet naturally reveals which features significantly influence each prediction through its masking mechanism. This level of transparency fosters greater trust, especially in sensitive fields like agriculture where accurate and explainable decision support is crucial. The results showcased in Table 5 and Fig. 21, highlight the comparative performance of TabNet against Random Forest and other existing methods. Across all evaluation metrics such as accuracy, precision, recall, and F1-score, TabNet consistently delivers superior outcomes. This establishes TabNet not only as a high-performing model but also as a more transparent and insightful alternative, making it particularly well-suited for practical applications in agricultural decision-making systems.

The analysis of results indicates that the TabNet-based system for recommending fertilizers and crops is a robust and reliable asset for precision agriculture. By integrating deep learning with an attention-driven approach, TabNet successfully highlights the most relevant features, enabling it to make smarter and more precise decisions. A key advantage of this system is its capacity to explain the reasoning behind its predictions. With built-in interpretability and support from SHAP analysis, users can easily identify which factors most significantly influenced a recommendation. This level of transparency fosters trust in the system's suggestions, empowering farmers and agronomists to make well-informed, dependable choices.

Conclusion

This study introduces an innovative and independent fertilizer and crop recommendation system that leverages deep learning, specifically the TabNet architecture, while incorporating SHAP for enhanced interpretability. The model examines crucial soil and environmental factors such as nutrient levels, pH, and climatic conditions to provide accurate, data-driven recommendations. Impressively, it achieved an accuracy rate of 99.65 percent for fertilizer suggestions and 96.21 percent for crop recommendations, significantly outperforming traditional classifiers like Random Forest. These results highlight the model's capacity to learn complex patterns and deliver reliable predictions, which are essential for smart farming initiatives. A key strength of the system is its focus on interpretability. The built-in attention mechanism of TabNet, combined with SHAP visualizations, allows users to understand which factors significantly influence each recommendation. This level of transparency builds trust among users and aids both farmers and agricultural experts in making informed decisions. Additionally,

techniques such as SMOTE for data balancing, iterative imputation, and fivefold cross-validation have notably enhanced the model's reliability and adaptability across different conditions.

Looking ahead, there is a plan to expand the system to cover a wider range of geographical areas and environmental scenarios. Integrating real-time data from IoT sensors for soil and crop monitoring could enable dynamic recommendations at the field level. Adapting TabNet or exploring lighter alternatives could also enhance accessibility and scalability in areas with limited resources. However, some limitations of the TabNet model warrant attention. It can be computationally intensive and often requires GPU acceleration, which may pose challenges for deployment in rural or edge environments. Additionally, the potential for misclassifications exists when fertilizers or crops share similar characteristics, particularly in less represented categories. Since the dataset used primarily focuses on Western Maharashtra, this may restrict the model's ability to generalize to other regions. Addressing these issues through broader datasets and optimization will be crucial for enhancing real-world effectiveness.

Data availability

This study employed two publicly available datasets. The Crop and Fertilizer Dataset for Western Maharashtra can be accessed via Kaggle at (<https://www.kaggle.com/datasets/sanchitaghola/crop-and-fertilizer-dataset-for-western-maharashtra>). The Crop Recommendation Dataset is available at (<https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset>). Both datasets are openly accessible and may be used for research purposes.

Code availability

No custom code was generated for this study. All analyses were conducted using established and publicly available Python libraries.

Received: 18 April 2025; Accepted: 31 October 2025

Published online: 24 November 2025

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Acknowledgements

I sincerely thank the editor and reviewers for their valuable suggestions and constructive feedback, which have greatly helped to improve the clarity and quality of this manuscript.

Author contributions

Stella Mary V: Conceptualization, Formal analysis, Investigation, Methodology, Software, Visualization, Writing—original draft, Writing—review & editing. Jayashree P: Conceptualization, Supervision, Proof reading, Review.

Funding

No funding was received.

Declarations

Competing interests

The authors declare no competing interests.

Ethics approval and consent to participate

Not applicable. This study used publicly available datasets and did not involve any experiments with human participants or animals.

Consent for publication

Not applicable.

Additional information

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